

RAHUL KUMAR

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DOB: 15-Jul-2003 [in](#) in/rahul-kundelwalli [github](#) https://github.com/rahulkundelwalli

EDUCATION

Course	Institute	Year of Passing	Result
B.Tech (CSE)	IIT Jammu	Expected Aug 2025	7.92 CGPA (Till 5th Semester)
12th Standard	STEP BY STEP PUBLIC SCHOOL	2020	81%
10th Standard	STEP BY STEP PUBLIC SCHOOL	2018	85.8%

WORK EXPERIENCE

IIT Jodhpur

Research Intern, Remote (May 2024 - Present)

- Conducting research on Federated Learning for skin disease segmentation under Dr. Bikash Santra at IIT Jodhpur.
- Implementing TransUNet to enhance the segmentation accuracy of skin disease images.
- Focusing on leveraging Federated Learning to improve data privacy and model robustness in medical image analysis.

Central Road Research Institute

Research Intern, New Delhi (Mar 2024 - Present)

- Spearheaded a collaborative research project to develop an algorithm for predicting California Bearing Ratio (CBR) of soils.
- Improved efficiency and accuracy in pavement design and construction projects by enhancing the developed algorithm.
- Utilized artificial intelligence techniques, including artificial neural networks and GBR, to refine the algorithm.
- Streamlined highway construction processes by identifying crucial parameters for CBR prediction models.

PROJECTS

Web Element Recognition Model | Python, YOLO, CNN [🔗](#)

IIT Jammu

- Created YOLOv8 Web Element Recognition Model for precise webpage element tagging using advanced computer vision.
- Utilized advanced computer vision techniques for precise identification and tagging of diverse webpage elements
- Implemented customizable training features for adaptation to specific use cases and domains
- Facilitated seamless integration of YOLOv8 with existing web-related pipelines
- Enhanced automation and efficiency in web tasks through accurate element detection and bounding box generation
- Optimized web scraping and monitoring processes with YOLOv8's element recognition capabilities

Crop Pest Detection: Real-Time Monitoring [🔗](#)

IIT Jammu

- Led the development of a real-time crop pest detection system
- Utilized the pre-trained EfficientNetV2 model for robust detection of crop pests
- Achieved live monitoring capabilities for timely intervention in agricultural settings
- Trained the model on Kaggle's Agricultural Pests Dataset to recognize various pests such as ants, bees, beetles, and caterpillars
- Orchestrated a live monitoring script using OpenCV for analysis of webcam feed
- Facilitated seamless integration into pest management workflows for enhanced agricultural productivity
- Utilized Python and libraries including TensorFlow, OpenCV, and scikit-learn for model training, inference, and live monitoring

Flower Classification | Matplotlib, TensorFlow, NumPy, Scikit-Learn, Keras [🔗](#)

IIT Jammu

- Successfully implemented a pretrained **MobileNetV2** model with an image input size of 224x224 to analyze 104 different flower classes on a large dataset utilizing Tensorflow and Scikit-Learn
- Optimized model parameters to facilitate greater accuracy and success in flower classifications

SKILLS

Programming Languages	C/C++, Python
Softwares	Git/Github, Jupyter Notebook
Machine Learning	Scikit-Learn, Numpy, Pandas, Matplotlib, Tensorflow , CNN, Deep Learning, YOLO, RNN
DataBase	MYSQL, PL/SQL

Coursework

Data Structures and Algorithm, Object Oriented Programming, Operating System , Data Base Management System

ADDITIONAL DETAILS

Professional Development

- Supervised Machine Learning: Regression and Classification (Coursera)
- Advanced Learning Algorithms: Neural Networks and Decision Trees (Coursera)

Workshops and Training Attended

- The Role of Data, Analytics & AI in Finance Transformation by Steve Rosvold
- Successfully completed TCS iON Career Edge - Young Professional

Positions of Responsibility

- Organized 'Modi@20' event in 2022
- Class Representative of CSE.
- Event Head/Host for Prom Night and TwinsFlame, Renao'23

Extracurricular Activities

- YouTube Content Creator with a channel boasting 27.4k subscribers

Hobbies: Badminton, Football, Gaming